

## 13.IMPLEMENT THE CAR PRICE PREDICTION MODEL USING PYTHON

### 1. Car Price Prediction – Linear Regression

| Engine | Age | Price |  |  |  |
|--------|-----|-------|--|--|--|
| 1000   | 8   | 3.0   |  |  |  |
| 1200   | 7   | 3.8   |  |  |  |
| 1500   | 5   | 5.0   |  |  |  |
| 1800   | 3   | 6.5   |  |  |  |
| 2000   | 2   | 7.5   |  |  |  |

#### PROGRAM

```
import pandas as pd
from sklearn.linear_model import LinearRegression
df = pd.read_csv("car1.csv")
X = df[["Engine", "Age"]]
y = df["Price"]
model = LinearRegression()
model.fit(X, y)
sample = [[1600, 4]]
print("Predicted Car Price:",
      round(model.predict(sample)[0], 2), "Lakhs")
```

#### OUTPUT

Predicted Car Price: 5.7 Lakhs

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## 2. Car Price Prediction – Random Forest

| Year | KM Driven | Price |
|------|-----------|-------|
| 2015 | 80000     | 4.0   |
| 2016 | 70000     | 4.5   |
| 2018 | 50000     | 6.0   |
| 2020 | 30000     | 8.0   |
| 2022 | 15000     | 10.0  |

### PROGRAM

```
import pandas as pd
from sklearn.ensemble import RandomForestRegressor
df = pd.read_csv("car2.csv")
X = df[["Year", "KM Driven"]]
y = df["Price"]
model = RandomForestRegressor(
    n_estimators=50, random_state=1)
model.fit(X, y)
sample = [[2019, 40000]]
print("Predicted Price:",
      round(model.predict(sample)[0], 2), "Lakhs")
```

### OUTPUT

Predicted Price: 7.0 Lakhs

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## 3. Car Price Prediction – Decision Tree

| <b>Mileage</b> | <b>Engine</b> | <b>Price</b> |
|----------------|---------------|--------------|
| 20             | 1000          | 4.0          |
| 18             | 1200          | 4.8          |
| 15             | 1500          | 6.0          |
| 12             | 1800          | 7.5          |
| 10             | 2000          | 9.0          |

### **PROGRAM**

```
import pandas as pd
from sklearn.tree import DecisionTreeRegressor
df = pd.read_csv("car3.csv")
X = df[["Mileage", "Engine"]]
y = df["Price"]
model = DecisionTreeRegressor(random_state=2)
model.fit(X, y)
Sample = [[14, 1600]]
print("Predicted Price:",
      round(model.predict(sample)[0], 2), "Lakhs")
```

### **OUTPUT**

Predicted Price: 6.0 Lakhs

## **4. Car Price Prediction – Polynomial Regression**

| <b>Car Age</b> | <b>Price</b> |
|----------------|--------------|
| 1              | 10.0         |
| 2              | 8.5          |

|   |     |
|---|-----|
| 3 | 7.2 |
| 4 | 6.0 |
| 5 | 5.0 |
|   |     |

## PROGRAM

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
df = pd.read_csv("car4.csv")
X = df[["Car Age"]]
y = df["Price"]
poly = PolynomialFeatures(degree=2)
Xp = poly.fit_transform(X)
model = LinearRegression()
model.fit(Xp, y)
sample = [[3.5]]
result = model.predict(poly.transform(sample))
print("Predicted Price:",
      round(result[0], 2), "Lakhs")
```

## OUTPUT

Predicted Price: 6.55 Lakhs

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## 5. Car Price Prediction – Multiple Linear Regression

| Year | Engine | KM Driven | Price |
|------|--------|-----------|-------|
|------|--------|-----------|-------|

|      |      |       |     |
|------|------|-------|-----|
| 2016 | 1200 | 70000 | 4.0 |
| 2017 | 1400 | 60000 | 4.8 |
| 2018 | 1500 | 50000 | 5.5 |
| 2020 | 1800 | 30000 | 7.5 |
| 2022 | 2000 | 15000 | 9.5 |

## PROGRAM

```
import pandas as pd
from sklearn.linear_model import LinearRegression
df = pd.read_csv("car5.csv")
X = df[["Year", "Engine", "KM Driven"]]
y = df["Price"]
model = LinearRegression()
model.fit(X, y)
sample = [[2019, 1600, 40000]]
print("Predicted Price:",
      round(model.predict(sample)[0], 2), "Lakhs")
```

## OUTPUT

Predicted Price: 6.5 Lakhs